BIOCHEMISTRY CUMULATIVE EXAM

February 15, 1992

1) Some of the most important tools available to the molecular biologist are restriction enzymes. What is the enzymatic activity that restriction enzymes possess that makes them valuable? (10 points)

2) The micro-organisms that restriction enzymes are isolated from did not evolve them to make life easier for molecular biologists. What is the probable role these enzymes serve in these bacteria? (10 points)

3) Name one of three individuals who received the Nobel prize for the discovery of restriction enzymes. (1 point)

4) There are two common forms of nucleic acid, DNA and RNA. Describe briefly the two key *chemical* differences between them. (i.e., How does DNA differ in covalent structure from RNA?) (10 points)

5) Another important tool in molecular biology is the ability to rapidly and accurately sequence DNA. In the late 70's two different methods for sequencing were published. One of these methods, Maxim-Gilbert, is no longer widely used. The other, Sanger or, as it is sometimes called, dideoxy sequencing, is the prevalent technique. Assume you have a single stranded DNA template you wish to sequence. Describe the basic steps used in Sanger sequencing up to, but not including, loading the gel. In particular, how is chain termination achieved? (35 points)

6) Depicted below is a pattern typical of an autoradiogram taken from a Sanger gel. (In practice, gels are often messier and more ambiguous than this representation.) Read the gel and write the DNA sequence (10 points)